

Answer Key — Biology Grade 11

Sub-Strand 3.2: Growth and Development in Animals • Lesson 3: Growth Curves: Why Do Vertebrates and Soft-Bodied Animals Grow Smoothly in an S-Shape While Insects with Hard Shells Grow in Steps?

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
A	D	B	C	B	C	D

USE AFTER: PREDICT PHASE — Predict: sketch of a broiler chick's mass over 10 weeks and 'Does a chick have an exuvia?'

Q1. A Kenyan broiler chick has no rigid outer shell like a grasshopper's exoskeleton. What does this suggest about a graph of the chick's mass against time over 10 weeks?

- A. A smooth, continuous curve with no sudden jumps**
- B. A staircase with flat parts and sudden jumps
- C. A straight line rising at the same rate every week
- D. A flat line because the chick only changes after moulting

Correct answer: A. Without a hard, non-stretching shell, the chick's cells and tissues increase continuously, so its mass graph is a smooth curve. A staircase is typical of moulting insects, and a straight line ignores the changing growth rate.

USE AFTER: OBSERVE PHASE — Observe: calculating growth rate for the exponential interval of Data Set A (chick mass)

Q2. A broiler chick from a Kiambu farm has a mass of 900 g at week 5 and 1500 g at week 7. Using Growth rate = (final value - initial value) / time interval, what is its growth rate between week 5 and week 7?

- A. 600 g/week
- B. 200 g/week
- C. 750 g/week
- D. 300 g/week**

Correct answer: D. Growth rate = $(1500 - 900) / 2 = 300$ g/week. 600 is the total gain without dividing by time, and the other values come from dividing by the wrong number or using the wrong masses.

USE AFTER: OBSERVE PHASE — Observe: Data Set C, grasshopper nymph body length with moult days marked

Q3. A grasshopper nymph's body length stays at 12 mm from day 10 to day 19. On day 20 it moults and its length jumps to 16 mm. Which explanation fits this stepped pattern?

- A. The nymph melts and rebuilds its whole body at each moult
- B. The hard exoskeleton cannot stretch, so body size increases only after the old one is shed**
- C. The old skin peels off like sunburned skin, so the nymph is always growing underneath
- D. The nymph stops feeding between moults, so it cannot grow

Correct answer: B. The rigid exoskeleton cannot stretch, so the graph is flat between moults. Size increases when the nymph sheds the old exoskeleton and the new one expands, giving a staircase graph.

USE AFTER: EXPLAIN PHASE — Explain: CER response on why the chick's graph forms a smooth S-shape

Q4. A chick's growth rate is 240 g/week in weeks 3-6, 80 g/week in weeks 7-8 and about 3 g/week in weeks 9-10. Which is the best biological reason for growth slowing and then almost stopping?

- A. Cells divide faster and faster as the chick gets older
- B. The chick develops a hard exoskeleton that cannot stretch
- C. Resources and genetic limits restrict growth as the chick nears its adult size**
- D. The chick sheds its skin, which removes mass

Correct answer: C. In the deceleration and stationary phases, food, space and genetic limits on adult size slow growth. Cell division does not keep speeding up, and chicks have no exoskeleton to moult.

USE AFTER: DQB PHASE — Driving Question Board: moving answered questions to the 'Answered' section with evidence cited

Q5. On the Driving Question Board, the question 'Do all animals grow the same way?' is to be moved to 'Answered'. Which evidence best supports moving it?

- A. Both a chick and a grasshopper get bigger as they get older
- B. A chick's mass graph is a smooth S-curve, while a grasshopper's length graph is a staircase with flat parts between moults**
- C. A chick and a grasshopper both start life very small
- D. A chick's graph and a grasshopper's graph both have labelled axes

Correct answer: B. The question is answered by showing that different animals have different growth patterns: continuous sigmoid growth in the chick and intermittent stepped growth in the insect. Getting bigger with age is true of both, so it does not show a difference.

USE AFTER: MODEL BUILDING PHASE — Model: revising the class model with a smooth S-curve labelled with lag, exponential, deceleration and stationary phases

Q6. Which list gives the four phases of a sigmoid (S-shaped) growth curve in the correct order from birth to adult size?

- A. Lag, deceleration, exponential, stationary
- B. Stationary, lag, exponential, deceleration
- C. Lag, exponential, deceleration, stationary**
- D. Exponential, lag, stationary, deceleration

Correct answer: C. Growth starts slowly (lag), then speeds up with rapid cell division (exponential), then slows as limits act (deceleration), and finally levels off at adult size (stationary).

USE AFTER: END OF LESSON — End: exit ticket question 2 (calf growth rate)

Q7. A Friesian calf from Nyahururu rises from 40 kg to 70 kg in 10 weeks. What is its average growth rate?

- A. 30 kg/week
- B. 11 kg/week
- C. 0.3 kg/week
- D. 3 kg/week**

Correct answer: D. Growth rate = $(70 - 40) / 10 = 3$ kg/week. 30 kg is the total gain without dividing by time, and 11 is the result of adding the masses and dividing by 10.